Coordinate Geometry

ANSWER KEY

Section 1: Lines, Lengths & Midpoints

1 Find the equation of the line passing through $A(2,\ -3)$ and $B(6, 5)$ in the form $y = mx + c$.

a. $y = 2x - 7$ ✓

b. $y = 2x + 1$

c. $y = \frac{1}{2}x - 4$

d. $y = 2x - 3$

Slope $m = \frac{5-(-3)}{6-2} = \frac{8}{4} = 2$. Line: $y - 5 = 2(x - 6) \Rightarrow y = 2x - 7$.

2 Find the length of the line segment joining $P(-1, 4)$ and $Q(5,\ -4)$.

a. **10** ✓

b. 8

c. $2\sqrt{7}$

d. 12

$d = \sqrt{(5-(-1))^2 + (-4-4)^2} = \sqrt{6^2 + (-8)^2} = \sqrt{36 + 64} = 10$.

3 Find the midpoint of the segment connecting $C(-3, 8)$ and $D(7,\ -2)$.

a. $(2, 3)$ ✓

b. $(5, 3)$

c. $(2, 5)$

d. $(4, 6)$

$M = \left(\frac{-3+7}{2}, \frac{8+(-2)}{2}\right) = (2, 3)$.

Section 2: Perpendicular Bisectors & Gradients

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4 Find the equation of the perpendicular bisector of the segment joining $A(1, 2)$ and $B(5, 10)$.

a. $x + 2y = 15$ ✓

b. $2x - y = 0$

c. $x - 2y = -9$

d. $2x + y = 12$

Midpoint $M = (3, 6)$. Gradient of $AB = \frac{10-2}{5-1} = 2$. Perpendicular gradient $m_\perp = -1/2$. Equation: $y - 6 = -\frac{1}{2}(x - 3) \Rightarrow 2y - 12 = -x + 3 \Rightarrow x + 2y = 15$.

5 The line L_1 has equation $3x - 4y = 12$. Find the equation of the line L_2 perpendicular to L_1 passing through $(2, 1)$.

a. $4x + 3y = 11$ ✓

b. $3x + 4y = 10$

c. $4x - 3y = 5$

d. $3x - 4y = 2$

Slope of L_1 is $3/4$. Perpendicular slope is $-4/3$. Equation: $y - 1 = -\frac{4}{3}(x - 2) \Rightarrow 3y - 3 = -4x + 8 \Rightarrow 4x + 3y = 11$.

6 Three points are given: $P(1, 1)$, $Q(4, 5)$, and $R(k, 9)$. If P, Q, R are collinear, find k.

a. **7** ✓

b. 6

c. 8

d. 5

Gradient $PQ = \frac{5-1}{4-1} = \frac{4}{3}$. Gradient $QR = \frac{9-5}{k-4} = \frac{4}{k-4} = \frac{4}{3} \Rightarrow k - 4 = 3 \Rightarrow k = 7$.

Section 3: Circles — Equations, Tangents & Normals

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7 Find the center and radius of the circle given by $x^2 + y^2 - 6x + 8y - 11 = 0$.

a. **Center** $(3, -4)$**, radius 6** ✓

b. Center $(-3, 4)$, radius 6

c. Center $(3, -4)$, radius $\sqrt{11}$

d. Center $(-3, 4)$, radius 36

$(x-3)^2 - 9 + (y+4)^2 - 16 - 11 = 0 \Rightarrow (x-3)^2 + (y+4)^2 = 36$. Center is $(3, -4)$, radius is $\sqrt{36} = 6$.

8 Find the equation of the tangent to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$.

a. $3x + 4y = 25$ ✓

b. $4x - 3y = 0$

c. $3x - 4y = 25$

d. $4x + 3y = 24$

The radius to $(3, 4)$ has gradient $4/3$. Tangent is perpendicular, with gradient $-3/4$. Equation: $y - 4 = -\frac{3}{4}(x-3) \Rightarrow 4y - 16 = -3x + 9 \Rightarrow 3x + 4y = 25$.

9 A circle has diameter with endpoints $A(-2, 1)$ and $B(4, 9)$. Find the equation of the circle.

a. $(x-1)^2 + (y-5)^2 = 25$ ✓

b. $(x+1)^2 + (y+5)^2 = 25$

c. $(x-1)^2 + (y-5)^2 = 100$

d. $(x-3)^2 + (y-4)^2 = 25$

Center $M = \left(\frac{-2+4}{2}, \frac{1+9}{2}\right) = (1, 5)$. Radius $r = \frac{1}{2}\sqrt{(4-(-2))^2 + (9-1)^2} = \frac{1}{2}\sqrt{36+64} = 5$. Equation: $(x-1)^2 + (y-5)^2 = 25$.

Section 4: Intersections & Exam-Standard Geometry

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10 Determine the relationship between the line $y = x - 5$ and the circle $x^2 + y^2 = 9$.

a. **The line does not intersect the circle ✓**

b. The line is tangent to the circle

c. The line intersects the circle in two distinct points

d. The line passes through the center of the circle

Substitute: $x^2 + (x-5)^2 = 9 \Rightarrow 2x^2 - 10x + 16 = 0 \Rightarrow x^2 - 5x + 8 = 0$. $\Delta = 25 - 32 = -7 < 0$. No real solutions, so no intersection.

11 Find the area of the triangle with vertices $A(0,0)$, $B(6,2)$, and $C(2,8)$.

a. **22 ✓**

b. 20

c. 24

d. 18

Area $= \frac{1}{2}|x_A(y_B - y_C) + x_B(y_C - y_A) + x_C(y_A - y_B)| = \frac{1}{2}|0 + 6(8-0) + 2(0-2)| = \frac{1}{2}|48 - 4| = 22$.

12 Find the length of the chord cut off from the line $y = 2$ by the circle $x^2 + y^2 = 13$.

a. **6 ✓**

b. 3

c. $2\sqrt{13}$

d. 8

Set $y = 2$: $x^2 + 4 = 13 \Rightarrow x^2 = 9 \Rightarrow x = \pm 3$. The points are $(-3, 2)$ and $(3, 2)$, so chord length is $3 - (-3) = 6$.

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